

MudWatt Bioelectric League

Can your team make mud generate electricity and defend your design with data?

Win condition: Highest combined score from peak reading, daily data quality, controlled-variable design, redesign notes, and final explanation.

THE SCIENCE YOU ARE USING

Electrogenic bacteria. Some bacteria already in ordinary soil are electrogenic: as they digest nutrients, they push spare electrons onto surfaces outside their cells. Bury a graphite anode in oxygen-free mud and these microbes coat it, feeding it a steady stream of electrons.

Completing the circuit. Electrons leave the anode, run through the wire, the 100 k Ω resistor, and the multimeter, reaching the cathode in the air on top, where they join oxygen and protons to form water. More microbes mean a higher voltage on the multimeter. Power is roughly voltage times current.

HOW TO BUILD AND RUN IT

- 01 Inoculate the mud.** Gloves on. Pack the vessel with sponge-damp soil. The bacteria you need are already in the dirt, so no culture is required.
- 02 Place the electrodes.** Bury the anode near the bottom and smooth mud over it so no air reaches it. Rest the cathode flat on top, exposed to the air.
- 03 Add a measured fuel.** Mix in a small, measured amount of one approved fuel. Record exactly how much. This is the food that drives the voltage.
- 04 Connect and read a baseline.** Clip the dry felt tabs to a 100 k Ω resistor, set the multimeter to DC millivolts, and read the voltage across the resistor. Day one is often weak; the biofilm needs time.
- 05 Control one variable.** Pick ONE thing to test all week: fuel amount, moisture, electrode contact area, or mud source. Change only that, predict the effect, and compare to baseline.

Safety: Gloves required. No food near mud station. Leave the lid off so the cathode stays in air. Wash hands after handling soil. Allergy: Do not use nut-containing food scraps. Preferred fuels are sugar water; optional collected leaf litter.

MATERIALS

Carbon felt electrodes (cut as a flag with a dry tab)	2 per cell
24 oz deli cups	1 per cell
100 k Ω resistor	1 per cell
Alligator leads	2 per cell
Fresh soil or mud	fill each vessel
Nitrile gloves	per student
Fuel	sugar water
Labels and tape	shared
Multimeters	required, 1 per team

YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

Prediction (what will work best, and why): _____

The ONE variable we are testing: _____

Baseline result: _____

After redesign: _____

DEFEND YOUR DESIGN

What evidence made you change your design? _____

If you ran it again, what is the ONE thing you would change? _____

HOW YOU SCORE (100 POINTS)

SCORING CRITERION	POINTS
Peak voltage (mV)	35
Daily data log quality	20
Controlled-variable design	15
Redesign or maintenance plan	15
Final explanation	15
Total	100